

Test 27: Binary chiral symmetry at $\nu=1/2$

Verdict: PASS | Generated: 2026-09-15 03:09:45 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

EXACT: defect 0 on pure lattice; $W>0$ row is the honest breaker control. ν -dictionary (v23.2 rework): pass 1 embeds 20,000 zeros (honest ν ? verdict), HARDCORE embeds the full

METHOD: EXACT: defect 0 on pure lattice; $W>0$ row is the honest breaker control. ν -dictionary (v23.2 rework): pass 1 embeds 20,000 zeros (honest ν ? verdict), HARDCORE em

Test 27 ν -DEEP (50k): pairing exact on the 100000-dim dictionary, ν ? 0.6119 (GUE 0.5996), Poisson control 0.3864, block spread 0.0050, KS D_GUE 0.0216 vs D_GOE 0.0881.

Test 27 HARDCORE: exact 0.00e+00, ν -row 5.77e-03 (ν _rms 0.0058), vortex 0.00e+00 (preserved), Coulomb W-scan 0.101?0.746 (ratio 7.42), ν -floor W-independent.

Test 27b [HARDCORE pass 2]: 8 sub-checks, 0 failed ν PASS

Result

- 'plots/plot_01.png' ν 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/plot_02.png' ν 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_03.png' ν 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ν animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 27 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[03:07:31.877] [+12042.906s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[03:07:42.466] [+12053.494s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=1.00, torus, 0 vortices, model=:monumental

[03:07:57.316] [+12069.144s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=0.00, torus, 2 vortices, model=:monumental

[03:08:09.555] [+12080.584s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=0.50, torus, 2 vortices, model=:monumental

[03:08:19.657] [+12090.685s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=1.00, torus, 2 vortices, model=:monumental

[03:08:33.394] [+12104.422s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=2.00, torus, 2 vortices, model=:monumental

[03:08:44.376] [+12115.405s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=4.00, torus, 2 vortices, model=:monumental

[03:08:58.626] [+12129.654s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=0.50, torus, 0 vortices, model=:monumental

[03:09:10.593] [+12141.621s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=1.00, torus, 0 vortices, model=:monumental

[03:09:21.957] [+12152.985s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=2.00, torus, 0 vortices, model=:monumental

[03:09:34.381] [+12165.410s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ν =0.5000, t=1.00, W=4.00, torus, 0 vortices, model=:monumental

[03:09:43.051] [+12174.079s] calc: KS D = 0.021619, p_asymptotic = 9.567870e-21 (n=49999, ν =4.8368)

[03:09:43.060] [+12174.089s] calc: KS D = 0.088053, p_asymptotic = 0.000000e+00 (n=49999, ν =19.6997)

CONSOLE CAPTURE

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TEST 27 (pass 2, HARDCORE): binary chiral symmetry ? deep audit

Exact row + ?_rms contract + vortex preservation + Coulomb W-scan

(launch-standard lattice, 96x96 in two-pass mode)

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? RUNTIME: 11 builds + 11 S•H•S defect evals at 96x96 (Diagonal S, O(N?)) + ?-50k deep statistics (O(n)) ? ? 2?4 min

H1 exact row (W=0, clean): defect = 0.00e+00 < ? 1e-10

H2 ? row (W=1.0, Nv=0): defect = 5.77e-03 vs ?_rms = 0.0058 (analytic) ? [0.5, 2.0]×?

H2v vortex row (Nv=2, W=0): defect = 0.00e+00 < 1e-10 ? (bond phases PRESERVE AIII)

W-scan (Nv=2) W=0.5: defect = 1.0052e-01

W-scan (Nv=2) W=1.0: defect = 2.0010e-01

W-scan (Nv=2) W=2.0: defect = 3.9425e-01

W-scan (Nv=2) W=4.0: defect = 7.4620e-01

H3 scan: monotone ?, defect(W=4)/defect(W=0.5) = 7.42 ? [5, 12] ?

Nv=0 ?-floor diagnostic (residual ? is FIXED-amplitude):

W-scan (Nv=0) W=0.5: defect = 5.7704e-03 ? ?_rms (W-independent ?)

W-scan (Nv=0) W=1.0: defect = 5.7704e-03 ? ?_rms (W-independent ?)

W-scan (Nv=0) W=2.0: defect = 5.7704e-03 ? ?_rms (W-independent ?)

W-scan (Nv=0) W=4.0: defect = 5.7704e-03 ? ?_rms (W-independent ?)

?-dictionary DEEP audit (pass 2) ? full 50000-zero embedded dataset:

Z1 monotone (50000 zeros): min ?? = 0.029538 ? OK

Z2 –E pairing (D=diag(–??/2), 100000-dim): max|d+d_?| = 0.0e+00, ?D+?D??/?D? = 0.00e+00, ???v?v? = 0.0e+00 ? EXACT AIII

Z3 zero modes of the dictionary: 0 (AIII index anchor; min|??| = 14.1347) ? OK

Z4 ?r?(49998 raw-gap ratios) = 0.6119 (GUE 0.5996; window –0.020) vs Poisson control 0.3864 (same n, separation 0.225 > 0.15) ? OK (GUE-class, discriminated)

Z5 block stationarity (5×10k): ?r? = 0.6147, 0.6121, 0.6120, 0.6097, 0.6109, spread = 0.0050 < 0.012 ? OK

Z6 KS: D_GUE = 0.021619 (< 0.030), D_GOE = 0.088053 ? GUE-class ?; depth diagnostic: no pass-1 baseline

??

exact row < 1e-10 PASS ? binary contract at ?=1/2, W=0: 0.00e+00

?_rms contract (W=1.0) PASS ? defect 5.77e-03 ? analytic 2???_F/?H?_F = 0.0058

vortices PRESERVE AIII PASS ? defect = 0.00e+00 (Nv=2, q=–1.00, W=0) ? bond phases; Byers-Yang for integer q

Coulomb W-scan linear PASS ? Nv=2: monotone, defect 0.101?0.746, ratio 7.42 ? [5, 12]

?-dict exact –E pairing PASS ? 100k-dim dictionary: max|d+d_?| = 0.0e+00, ?D+?D??/?D? = 0.00e+00, ?? probe 0.0e+00, tower 0

?-50k ?r? GUE-window + control PASS ? ?r? = 0.6119 (GUE 0.5996 –0.020), Poisson control 0.3864, separation 0.225 > 0.15

?-50k block stationarity PASS ? 5×10k blocks spread 0.0050 < 0.012

?-50k KS GUE-class PASS ? D_GUE = 0.0216 < 0.030 and < D_GOE = 0.0881 (Poisson would be ?0.28)

Test 27b [HARDCORE pass 2]: 8 sub-checks, 0 failed ? PASS